



It works the fault, not the keyword.

Most “AI for HVAC” is a chatbot that answers whatever you type. ThermoRivet runs a diagnostic engine: it holds every candidate cause at once, asks for the single test that best separates them, and updates the ranking from what you actually measured.

The reasoning is deterministic and runs on our own servers — the same evidence produces the same walk, every time, and every conclusion can be replayed step by step. That is what makes it defensible in front of a customer, and what a language model on its own cannot give you.

The rule the whole product is built on

Nothing expensive gets condemned on circumstantial evidence. A compressor, a heat exchanger or a control board requires a test that confirms it directly — no matter how far in front it is on the readings.

ON A CALL IT

- Takes the complaint in the words the customer used, or spoken aloud
- Asks the one test that best separates what is still on the table
- Reads the rating plate from a photo, and asks again if it cannot
- Scopes a fault code to the model and control board before interpreting it

AND THEN

- Names the repair with the parts, or says it does not have enough to name one
- Files the whole walk against the customer and the job
- Exports a service report the customer can read
- Keeps every reading with a note of where the number came from

1

question at a time

Cheapest informative test first, with the reason it was chosen. Never a checklist of forty readings.

41

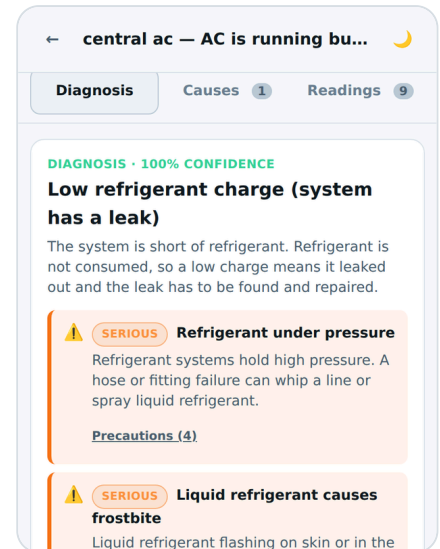
candidate causes

Ranked continuously across electrical, refrigeration, heating and airflow — with what rules each one out.

91

automated tests

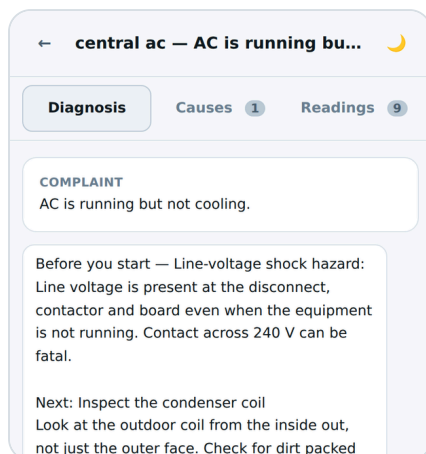
Including an evaluation suite that checks the *route* to an answer, not just the answer.



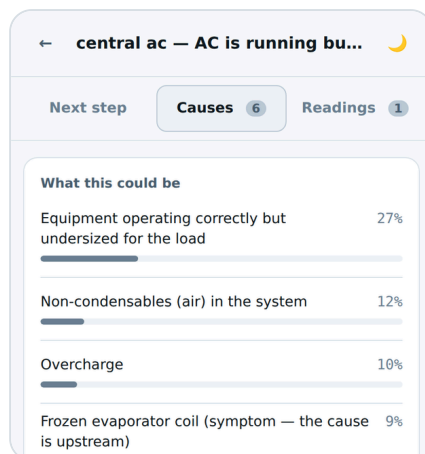
01 A finished diagnosis: the conclusion, the hazards that apply to it, and the evidence behind it.

Runs in any phone browser and installs to the home screen. Dark by default, 48 px touch targets, 16 px inputs — built for a phone held in a gloved hand in a mechanical room, not for a desk.

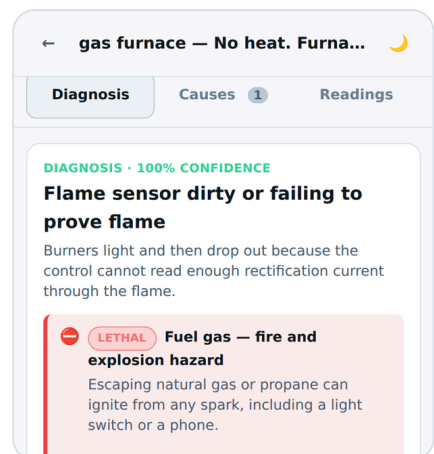
Evidence in, ranking out, conclusion last



02 One question. The hazard for the step is rendered above the instruction and a lethal one cannot be dismissed.



03 Live ranking. Every candidate with its current probability. Tap one to see the findings driving it — and the ones arguing against.



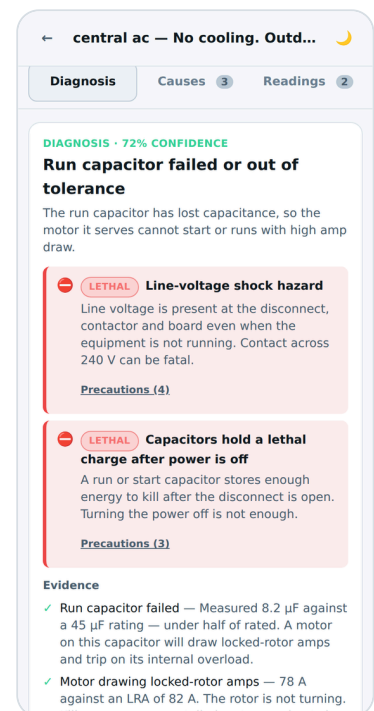
04 Conclusion. Flame proving localised in two questions, with the repair, the parts and every hazard that applies.

The reading that gets compressors replaced

A condenser humming and drawing **78 A against an 82 A locked-rotor rating** is a stalled motor. It is also the single most common reason a healthy compressor gets condemned and a customer gets a five-figure quote they did not need.

In our own recorded walk, “compressor mechanically seized” was leading the ranking one step earlier. The engine refused to conclude on it, called for the capacitor test, and found **8.2 μ F against a 45 μ F rating**. The answer was a \$30 part.

That behaviour is not a prompt asking the model to be careful. It is a property of the engine — an expensive hypothesis carries a required-evidence flag and cannot be concluded without a test that confirms it — and it has a test that fails if it ever regresses.



05 The same evidence, the correct part.

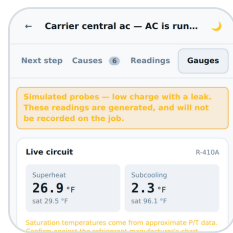
It says when it does not know

On a system where every reading is in range it reaches **no conclusion at all** rather than manufacturing a fault. An app that invents a failure on healthy equipment costs you the customer.

Fault codes scoped to the board

A Carrier code means different things on different boards. Without one, the app returns **every meaning it holds** and asks for the model number rather than picking. Entries stay marked unverified until confirmed against a manufacturer document.

The rest of the call



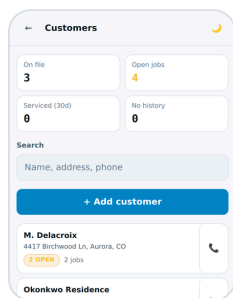
06 Live circuit. Amber banner marks the built-in simulator.

Wireless gauges, read live

Pair a wireless probe set and superheat and subcooling recompute on every packet — dew point on the low side, bubble point on the liquid line, so a zeotropic blend stays honest. Watching superheat settle is how you know the system has stabilised; a snapshot typed afterwards is a reading of the transient.

A clamp feeds nothing until you say where it is fitted, a reading over a minute old is refused, and two probes claiming the same line is reported rather than silently overwritten.

Working today: any probe using the published Bluetooth standard profile, on Android, Windows and macOS. **Not yet:** Fieldpiece JobLink, Testo and Yellow Jacket — see page 4.

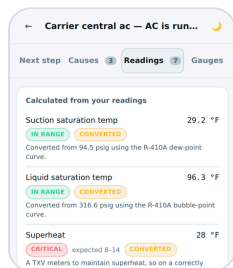


07 The book, sorted by who you saw last.

Customers and service history

Every job files itself against the customer. Open the site and you get the contact, the gate code and the dog, the equipment on site, and every diagnosis and report ever raised there — before you knock.

A repeat visit attaches to the existing record instead of creating a second copy of the same house. Call, text and directions are one tap from the customer screen. On a company plan the book is shared across the crew; a solo technician's records stay private to them.

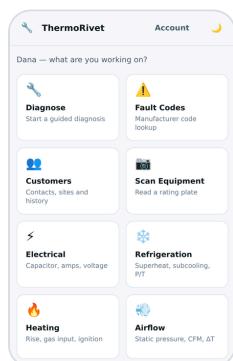


08 Every conversion says where it came from.

Numbers that carry their provenance

Superheat, subcooling, evaporator TD, temperature rise, static pressure, capacitor tolerance, gas input — each with its expected band and a plain-English reading of what it means. Anything converted from a pressure is tagged **CONVERTED** and tells you to confirm against the manufacturer's P/T chart before acting on a marginal value.

Model-number decoding separates what it actually read from what it inferred. Photo analysis of a rating plate never guesses a character it could not see — it asks for a better picture. Service reports export to PDF with the readings, the conclusion and what was ruled out.



09 One screen, everything on the truck.

The rest of the toolkit

Manufacturer fault-code lookup, rating-plate scanning, and the four calculator sets a technician reaches for most — electrical (capacitor tolerance, amp draw, voltage drop), refrigeration (P/T, superheat, subcooling, target charge), heating (temperature rise, gas input, ignition sequence) and airflow (static pressure, CFM, delta-T).

Each is the same engine's arithmetic, so a number worked out on a calculator screen means what it means inside a diagnosis.

Safety is not advisory

Line voltage, capacitor storage, fuel gas, refrigerant pressure, combustion and CO, arc flash and rotating equipment each carry a banner above the step they apply to. The app will not help bypass a safety control, and the guardrail runs on its own generated text as well as on what you type.

Built for the conditions

Dark by default because most service work happens in dark spaces. Severity is never carried by colour alone. Voice input runs on the phone itself. The full diagnostic engine works with no AI key configured at all — a language model only changes the wording, never the decision.

Priced per technician

Free

\$0

Try the engine

- 5 guided diagnoses a month
- Fault-code lookup
- Refrigerant P/T and superheat calculators

Pro

MOST TECHNICIANS

\$29 / month

or \$290 a year — two months free

- Unlimited guided diagnoses
- Wireless gauge readings
- Photo analysis and model decoding
- Full fault-code database
- Customers and service history
- Service reports with PDF export
- Voice input in the field

Company

\$79.99 / month

10 technician seats included

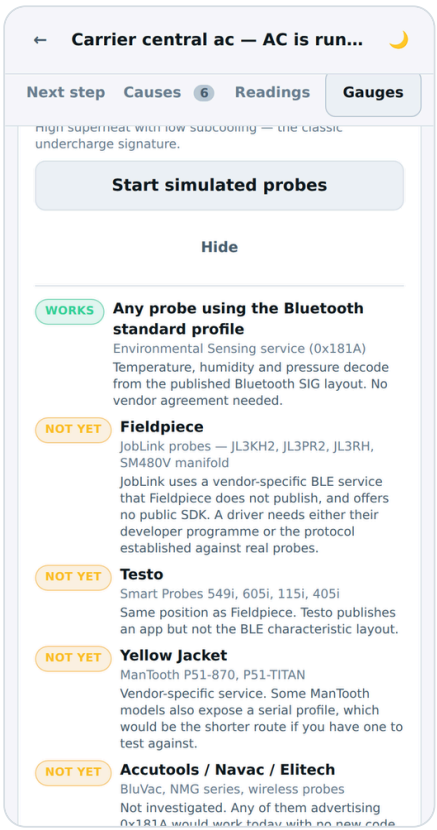
- Everything in Pro, for every technician
- Shared customer book across the crew
- Your own manuals in the knowledge base
- Branded service reports
- Priority support

Where it stands today

We would rather you knew this before you bought it than after. Everything on the previous three pages is built and running. These are the edges.

- Fieldpiece, Testo and Yellow Jacket probes do not connect yet

Their probes speak Bluetooth, but none of the three publishes the protocol and none offers a public SDK. A driver written from guesswork would turn arbitrary bytes into plausible-looking pressures that a technician then adjusts a charge from — a silent failure worse than no wireless support. So we have not shipped one. Probes using the published Bluetooth standard profile work now, and the app lists the exact status per vendor on the pairing screen.
- iOS wireless probes need the native app.** Apple has never shipped Web Bluetooth, and every iOS browser is WebKit underneath. Everything else works in Mobile Safari today.
 - The iOS app is not on the App Store yet.** Account deletion, privacy and support pages, and Apple In-App Purchase are all built and ready for review.
 - Boilers, hydronics and VRF are thin.** The engine's depth is in split systems, packaged units, gas furnaces and heat pumps.
 - Fault-code data is marked unverified** until confirmed against a manufacturer document. The app says so on every entry rather than pretending otherwise.



10 The support list is in the app, on the pairing screen — not buried in a footnote.

COMPANY	EMAIL	PHONE	WEB
Your Company Name	you@yourdomain.com	(000) 000-0000	thermorivet.example.com

ThermoRivet assists a qualified technician. It does not replace one. Always follow applicable codes, the manufacturer's procedures, and your own safety practices. Screenshots are of the working application; readings shown against the built-in simulator are marked as such on screen.